

Question 1:

$$I_{D3} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{ov3})^2$$

$$50 \times 10^{-6} = \frac{1}{2} \times 50 \times 10^{-6} \times 50 \times (V_{ov3})^2$$

$$V_{ov3} = 0.2 \text{ V}$$

$$I_{D1} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{ov1})^2$$

$$50 \times 10^{-6} = \frac{1}{2} \times 100 \times 10^{-6} \times 50 \times (V_{ov1})^2$$

$$V_{ov1} = 0.14 \text{ V}$$

$$I_{D5} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{ov5})^2$$

$$100 \times 10^{-6} = \frac{1}{2} \times 100 \times 10^{-6} \times 50 \times (V_{ov5})^2$$

$$V_{ov5} = 0.2 \text{ V}$$

$$A_d = g_{m1} (r_{o1} \parallel r_{o3} \parallel R)$$

$$g_{m1} = \frac{2I_{D1}}{V_{ov1}} = \frac{2 \times 50 \times 10^{-6}}{0.14} = 0.714 \text{ mS}$$

$$r_{o1} = \frac{1}{\lambda_n I_{D1}} = \frac{1}{0.1 \times 50 \times 10^{-6}} = 200 \text{ k}\Omega$$

$$r_{o3} = \frac{1}{\lambda_p I_{D3}} = \frac{1}{0.2 \times 50 \times 10^{-6}} = 100 \text{ k}\Omega$$

$$\therefore A_d = 0.714 \text{ mS} (200 \text{ k} \parallel 100 \text{ k} \parallel 10 \text{ M}\Omega)$$

$$= 47.28 \text{ V/V}$$

2

$$A_{cm} = \frac{r_{o3}}{2r_{o5} + \frac{1}{g_{m1}}} \approx \frac{r_{o3}}{2r_{o5}}$$

$$r_{o5} = \frac{1}{\lambda_n I_{D5}} = \frac{1}{0.1 \times 100 \times 10^{-6}} = 100 \text{ k}\Omega$$

$$\therefore A_{cm} = \frac{100 \text{ k}}{2 \times 100 \text{ k}} = 0.5 \text{ V/V}$$

2

$$V_{ocm, \max} = V_{DD} - |V_{ov3}| = 1.8 - 0.2 = 1.6 \text{ V}$$

$$V_{ocm, \min} = V_{ov5} + V_{ov1} = 0.2 + 0.14 = 0.34 \text{ V}$$

$$V_{ocm, \text{opt}} = 0.97 \text{ V}$$

2

$$V_{in, \text{cm}, \min} = V_{ov5} + V_{ov1} + V_{th1} = 0.2 + 0.14 + 0.4 = 0.74 \text{ V}$$

$$V_{in, \text{cm}, \max} = V_{ocm, \text{opt}} + V_{th1} = 0.97 + 0.4 = 1.37 \text{ V}$$

$$\text{Expected value of common mode gain} = \frac{2 \times 0.97 \times \frac{1}{100}}{1.37 - 0.74} = \frac{0.0194}{0.63} = 0.031 \text{ V} \rightarrow 3$$

$$\text{CMRR without common mode feedback} = \frac{A_d}{A_{cm}} = \frac{47.28}{0.5} = 94.56 \rightarrow 3$$

$$\text{Gain of feedback amplifier} = g_{m8} \left(\frac{1}{g_{m10}} \parallel r_{o10} \parallel r_{o8} \right) / 2$$

$$\approx \frac{g_{m8}}{2g_{m10}}$$

$$g_{m8} = \sqrt{2\mu_p C_{ox} \frac{W}{L} I_D}$$

$$= \sqrt{2 \times 50 \times 10^{-6} \times 50 \times 100 \times 10^{-6}}$$

$$= 0.707 \text{ mS}$$

$$g_{m10} = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}$$

$$= \sqrt{2 \times 100 \times 10^{-6} \times 50 \times 100 \times 10^{-6}}$$

$$= 0.316 \text{ mS}$$

$$\therefore \frac{V_E}{V_{OCM}} = \frac{0.707}{2 \times 0.316} = 1.118 \text{ V/V}$$

$$\frac{V_r}{V_E} = \frac{g_{m5}}{2} \left[r_{o4} \parallel (g_{m2} r_{o2} 2r_{o5}) \right]$$

$$g_{m5} = \frac{2I_D}{V_{OV}} = \frac{2 \times 100 \times 10^{-6}}{0.2} = 1 \text{ mS}$$

$$\therefore \frac{V_r}{V_E} = \frac{1 \times 10^{-3}}{2} \left[100 \text{K} \parallel (0.714 \text{ mS} \times 200 \text{K} \times 2 \times 100 \text{K}) \right]$$

$$= 0.5 \times 10^{-3} [100 \text{K} \parallel 28.56 \text{M}]$$

$$= 49.82 \approx 50$$

$$\therefore \text{loop gain} = 1.118 \times 49.82 = 55.7$$

$$\therefore A_{cmf} = \frac{A_{cm}}{1 + \text{loop gain}} = \frac{0.5}{1 + 55.7} = 0.0088 \text{ V/V}$$

$$= 8.8 \times 10^{-3} \text{ V/V}$$

$$\therefore \text{CMRR} = \frac{47.28}{0.0088} = 5372.7$$

Question 2 :

$$\text{slew rate} = I_{D5}/C_c = \frac{20 \times 10^{-6}}{0.22 \times 10^{-12}} = 90.9 \text{ V}/\mu\text{s}$$

$$\text{Gain bandwidth product} = \frac{g_{m1}}{2\pi C_c}$$

$$g_{m1} = \sqrt{2\mu_p C_{ox} \frac{W}{L} I_D}$$

$$= \sqrt{2 \times 50 \times 10^{-6} \times 10 \times 10 \times 10^{-6}}$$

$$= 0.1 \text{ mS}$$

$$\therefore \text{GBP} = \frac{0.1 \times 10^{-3}}{2\pi \times 0.22 \times 10^{-12}} = 72.34 \text{ MHz}$$

$$\text{DC gain} = g_{m1} (r_{o2} \parallel r_{o4}) \times g_{m7} (r_{o7} \parallel r_{o8})$$

$$I_{D3} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$$

$$10 \times 10^{-6} = \frac{1}{2} \times 100 \times 10^{-6} \times 10 \times (V_{GS} - 0.4)^2$$

$$V_{GS3} = V_{GS4} = 0.54 \text{ V} = V_{GS}$$

$$\frac{I_{D7}}{I_{D4}} = \frac{(W/L)_7}{(W/L)_4} = \frac{80 \times 10^{-6}}{10 \times 10^{-6}}$$

$$\therefore I_{D7} = 8 \times 10 \times 10^{-6} = 80 \mu\text{A}$$

$$g_{m7} = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})$$

$$= 100 \times 10^{-6} \times 80 \times (0.54 - 0.4)$$

$$= 1.12 \text{ mS}$$

$$r_{o2} = \frac{1}{\lambda_p I_D} = \frac{1}{0.2 \times 10 \times 10^{-6}} = 500 \text{ k}\Omega$$

$$r_{o4} = \frac{1}{\lambda_n I_D} = \frac{1}{0.1 \times 10 \times 10^{-6}} = 1 \text{ M}\Omega$$

$$r_{o7} = \frac{1}{\lambda_n I_D} = \frac{1}{0.1 \times 80 \times 10^{-6}} = 125 \text{ k}\Omega$$

$$r_{o8} = \frac{1}{\lambda_p I_D} = \frac{1}{0.2 \times 80 \times 10^{-6}} = 62.5 \text{ k}\Omega$$

$$\therefore \text{DC gain} = 0.1 \times 10^3 (500 \text{ k} \parallel 1 \text{ M}) \times 1.12 \times 10^3 (125 \text{ k} \parallel 62.5 \text{ k})$$

$$= 33.33 \times 46.67 = 1555.3 \text{ V/V} \quad \text{--- 2}$$

$$\text{Bandwidth} = \frac{72.34 \times 10^6}{1555.3} = 46.5 \text{ kHz} \quad \text{--- 2}$$

$$\text{Output pole} = \frac{g_{m7}}{2\pi C_L} = \frac{1.12 \times 10^{-3}}{2\pi \times 1 \times 10^{-12}} = 178.25 \text{ MHz} \quad \text{--- 2}$$

$$\text{Zero} = \frac{g_{m7}}{2\pi C_C} = \frac{1.12 \times 10^{-3}}{2\pi \times 0.22 \times 10^{-12}} = 810.24 \text{ MHz} \quad \text{--- 3}$$

$$\therefore \text{phase margin} = 180 - \tan^{-1}(1555.3) - \tan^{-1}\left(\frac{75.34 \times 10^6}{178.25 \times 10^6}\right) - \tan^{-1}\left(\frac{75.34 \times 10^6}{810.24 \times 10^6}\right)$$

$$= 180 - 90^\circ - 22.91 - 5.31$$

$$= 61.78^\circ \quad \text{--- 3}$$

Question 3 :

$$\text{Gain of the amplifier} = g_{m2} [g_{m4} r_{o4} r_{o2} \parallel g_{m6} r_{o6} r_{o8}]$$

$$g_{m2} = \frac{2I_D}{V_{OV2}} = \frac{2 \times 25 \times 10^{-6}}{0.2} = 0.25 \text{ mS} = g_{m4}$$

$$g_{m6} = \frac{2I_D}{V_{OV6}} = \frac{2 \times 25 \times 10^{-6}}{0.1} = 0.5 \text{ mS}$$

$$r_{o4} = r_{o2} = \frac{1}{\lambda_p I_D} = \frac{1}{0.2 \times 25 \times 10^{-6}} = 200 \text{ k}\Omega$$

$$r_{o6} = r_{o8} = \frac{1}{\lambda_n I_D} = \frac{1}{0.1 \times 25 \times 10^{-6}} = 400 \text{ k}\Omega$$

$$\therefore \text{Gain} = 0.25 \times 10^{-3} [0.25 \text{ mS} \times 200 \text{ k} \times 200 \text{ k} \parallel 0.5 \text{ mS} \times 400 \text{ k} \times 400 \text{ k}]$$

$$= 0.25 \times 10^{-3} [10 \text{ M} \parallel 80 \text{ M}]$$

$$= 2222.22 \text{ V/V} \quad \text{--- 2}$$

$$V_{out, \max} = V_b + |V_{thp}| = 0.7 + 0.5 = 1.2 \text{ V} \quad \text{--- 2}$$

$$V_{out, \min} = V_{OV3} + V_{OV6} + V_{th,n} = 0.1 + 0.1 + 0.4 = 0.6 \text{ V} \quad \text{--- 2}$$

If V_{out} is shorted with V_{in2} :

For M_2 to be in saturation

$$V_{SD} > V_{SG} - |V_{thp}| \text{ for } M_2$$

$$V_S - V_D > V_S - V_G - |V_{thp}|$$

$$-V_D > -V_G - |V_{thp}|$$

$$V_D < V_G + |V_{thp}|$$

$$V_b + |V_{GS4}| < V_{out} + 0.5$$

$$0.7 + 0.7 < V_{out} + 0.5$$

$$V_{out} > 0.9$$

$$V_{out, \min} = 0.9 \text{ V} \quad \text{--- 2}$$

For M_4 to be in saturation

$$V_{SD} > V_{SG} - |V_{thp}|$$

$$V_S - V_D > V_S - V_G - |V_{thp}|$$

$$V_D < V_G + |V_{thp}|$$

$$V_{out} < V_b + |V_{thp}|$$

$$V_{out} < 0.7 + 0.5$$

$$V_{out} < 1.2$$

$$V_{out, \max} = \underline{1.2 \text{ V}} \quad \text{--- 2}$$

Question 4:

$$\text{Gain of auxiliary amplifier} = g_{m7} [g_{m5} r_{o5} r_{o6} \parallel g_{m4} r_{o4} (r_{o3} \parallel r_{o7})]$$

$$g_{m7} = \frac{2I_D}{V_{ov7}} = \frac{2 \times 10 \times 10^{-6}}{0.2} = 0.1 \text{ mS} = g_{m5}$$

$$g_{m4} = \frac{2I_D}{V_{ov4}} = \frac{2 \times 10 \times 10^{-6}}{0.1} = 0.2 \text{ mS}$$

$$r_{o5} = r_{o6} = r_{o7} = \frac{1}{\lambda_p I_D} = \frac{1}{0.2 \times 10 \times 10^{-6}} = 500 \text{ k}\Omega$$

$$r_{o4} = r_{o2} = r_{o1} = \frac{1}{\lambda_n I_D} = \frac{1}{0.1 \times 10 \times 10^{-6}} = 1 \text{ M}\Omega$$

$$r_{o3} = \frac{1}{\lambda_n I_D} = \frac{1}{0.1 \times 20 \times 10^{-6}} = 500 \text{ k}\Omega$$

$$\begin{aligned} \therefore A_{aux} &= 0.1 \text{ mS} [0.1 \text{ mS} \times 500 \text{ k} \times 500 \text{ k} \parallel 0.2 \text{ mS} \times 1 \text{ M} \times (500 \text{ k} \parallel 500 \text{ k})] \\ &= 0.1 \text{ mS} [25 \text{ M} \parallel 50 \text{ M}] \\ &= 1666.67 \text{ V/V} \end{aligned}$$

$$R_{out} = A_{aux} g_{m2} r_{o2} r_{o1}$$

$$g_{m2} = \frac{2I_D}{V_{ov2}} = \frac{2 \times 10 \times 10^{-6}}{0.1} = 0.2 \text{ mS}$$

$$\therefore R_{out} = 1666.67 \times 0.2 \times 10^{-3} \times 1 \text{ M} \times 1 \text{ M} = \underline{333.3 \text{ G}\Omega} \quad \text{--- 3}$$

$$V_{b2, \min} = V_{ov3} + V_{gs4} = 0.1 + 0.1 + 0.4 = 0.6 \text{ V} \quad \text{--- 2}$$

$$V_{b2, \max} = 1 \text{ V} \quad \text{--- 2}$$

Question 5

$K_n = K_p$ for symmetric inverter, $K_R = 1$, $V_{th} = 2.5 \text{ V}$

NMOS is in triode & PMOS in saturation

$$I_{Dn} = \frac{1}{2} K_n \left[2(V_{in} - 0.8)0.6 - 0.6^2 \right]$$

$$I_{Dp} = \frac{1}{2} K_p (V_{DD} - |V_{Tp}| - V_{in})^2$$

$$(4.2 - V_{in})^2 = 1.2 V_{in} - 1.32$$

$$V_{in} = 2.78 \text{ V} \quad \text{--- 6}$$

for INV 2

NMOS is in cutoff

$$\therefore V_{out2} = V_{DD} = 5 \text{ V} \quad \text{--- 2}$$